

DIFFUSION POSTERIOR SAMPLING FOR SLITLESS SPECTRAL IMAGING

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ABSTRACT

Imaging spectroscopy is a fundamental technique for investigating various physical phenomena. Traditional imaging spectrographs require a time-consuming scanning process to construct the spatial-spectral data cube, rendering them unsuitable for dynamic scenes. Slitless spectrographs have been proposed as snapshot imagers; however, their measurements consist of overlapped spectra, requiring to solve a limited-angle tomography problem. The existing reconstruction approaches have limited accuracy due to the highly ill-posed nature of the problem. In this work, we adapt the diffusion posterior sampling (DPS) method which uses a trained diffusion model as a prior. Targeting applications where spectrum consists of discrete emissions, we use a Gaussian parametrization for the spectrum, and use DPS to solve the resulting nonlinear inverse problem to estimate the Gaussian line parameters at each spatial location. Effectiveness of the approach is demonstrated on a solar imaging application.

Index Terms— Spectral imaging, nonlinear inverse problems, posterior sampling, diffusion models

1. INTRODUCTION

Spectral imaging, which captures images of a scene as a function of wavelength, is a fundamental diagnostic technique in physical sciences with widespread applications in agriculture [1], remote sensing [2, 3], astronomy [4], and medicine [5, 6]. For example, the shape and position of emission lines in space plasma spectra can reveal crucial properties like Doppler velocity, temperature, and density [7]. However, conventional spectral imagers are fundamentally limited by a slow mechanical scanning process needed to build a 3D spatial-spectral data cube from 2D detector measurements. This makes them unsuitable for studying dynamic scenes that evolve faster than the acquisition time [8].

To address this, snapshot spectral imagers that capture the full data cube in a single exposure have been developed [8–10]. Prominent architectures include the Coded Aperture Snapshot Spectral Imager (CASSI) [11, 12] and the Computed Tomography Imaging Spectrometer (CTIS) [13, 14]. CASSI systems use a coded aperture that, while enabling compressive reconstructions, adds optical complexity and

reduces photon throughput—a significant drawback in low signal-to-noise (SNR) applications. Conventional CTIS designs, on the other hand, use a 2D disperser to project a grid of diffraction orders onto the detector [15]. While this provides many tomographic projections, it requires a large detector area, forcing a trade-off that significantly restricts the instrument’s field of view.

In this work, we focus on an optically simpler variant, the multi-order Slitless Spectral Imager (SSI) [16]. An SSI uses a 1D disperser to capture only a few diffraction orders, which allows for a much larger instantaneous field of view on a conventional detector. This architectural trade-off—a larger field of view for fewer projections—makes the reconstruction an even more severely ill-posed inverse problem, for which existing methods have proven insufficient. Traditional non-parametric techniques like the Multiplicative Algebraic Reconstruction Technique (MART) tend to oversmooth spectral features and systematically underestimate key physical parameters [17]. While more recent parametric approaches [18, 19] reduce the number of unknowns by modeling the spectrum with a Gaussian profile, they formulate a challenging nonlinear and nonconvex problem. The solvers for these problems still typically rely on simple smoothness priors that cannot fully capture the underlying structure of the data.

We tackle the challenging reconstruction problem for SSIs, with an approach specifically designed for scenes characterized by discrete emission spectra. Such scenes are common in many scientific domains, from atmospheric physics [20] to the astrophysical imaging of space plasmas [7, 21]. We focus on the latter, targeting applications in solar physics where emission lines encode crucial plasma properties. Following a physically-motivated approach [18], we model these emission lines with Gaussian profiles. This parameterization transforms the task into a challenging nonlinear inverse problem of estimating the Gaussian parameters at each spatial location. To solve this, we adapt the Diffusion Posterior Sampling (DPS) algorithm [22], a framework particularly suited for such nonlinear problems as it can leverage a powerful, unconditionally trained generative model as a prior without requiring paired training data.

We demonstrate our method on the spectral imaging of the solar corona, focusing on a prominent emission line, where

the Gaussian parameters directly correspond to plasma intensity, Doppler velocity, and temperature [19]. The diffusion prior is trained on thousands of spectral scans from the EIS instrument on the Hinode observatory [23], and measurements are simulated for a well-established SSI design [17]. We show that our method not only achieves reliable reconstructions on test data but also generalizes better to out-of-distribution phantoms compared to a supervised end-to-end U-Net. The main contributions of our work are:

- A novel framework for slitless spectral imaging reconstruction by formulating it as a nonlinear inverse problem with a Gaussian parametric model.
- The adaptation of Diffusion Posterior Sampling (DPS) to solve this problem, using a score-based model as a prior for the physical line parameters.
- A demonstration of superior generalization performance against supervised deep learning methods on an out-of-distribution phantom, highlighting the robustness of our approach.

2. PROBLEM FORMULATION

The multi-order Slitless Spectral Imager (SSI) design considered in this work is based on the Multi-Order Solar EUV Spectrograph (MOSES) [16]. The instrument operates in a narrow EUV band containing a single spectral line with a Gaussian shape, characterized by three parameters: integrated line intensity, line width, and Doppler shift. These parameters vary over the field of view (FOV), and the goal is to estimate them at all spatial positions.

Let $a \in \{0, -1, 1, -2, 2, \dots\}$ denote the diffraction order, and consider K detectors placed at distinct diffraction orders $a_1, a_2, \dots, a_K$, each with $M \times M$ pixels. The relation between the spatial-spectral data cube $r(x, y, \lambda)$ and the detector measurements $y^{(a)}(x', y')$ can be expressed as line integrals similar to parallel beam tomography, where the projection angle θ is determined by the diffraction order a according to $\theta = \tan^{-1}(a)$. Assuming the dispersion direction is parallel to the x axis, the governing integral equation is:

$$y^{(a)}(x', y') = \int_{\lambda_0 - \delta}^{\lambda_0 + \delta} r(x' - a\lambda + a\lambda_0, y', \lambda) d\lambda \quad (1)$$

where the integration paths have slope $-a$ in the (x, λ) plane within the instrument bandpass of width 2δ centered around rest wavelength λ_0 . For simplicity we will write down the forward model equations for a fixed y' without loss of generality. Incorporating the Gaussian parametrization into Eq. (1) we obtain:

$$y^{(a)} = \sum_{m=1}^M f_m \mathcal{N}(\mathbf{x}' | a\mu_m + m, |a|\sigma_m) = \sum_{m=1}^M f_m \mathbf{h}_m^{(a)} \quad (2)$$

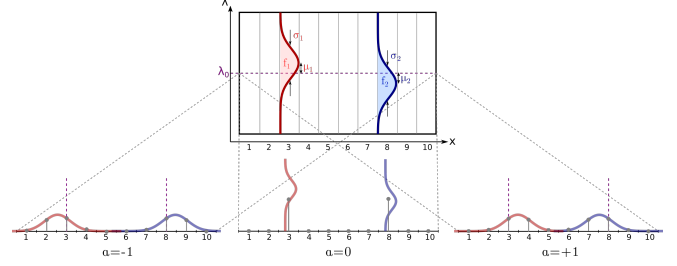


Fig. 1: Illustration of SSI forward model. Each diffraction order contains a projection of the spatial-spectral data cube.

where $\mathbf{y}^{(a)}$ is the vector holding the detector measurements at the diffraction order a for the pixel positions $\mathbf{x}' = [1 \ 2 \ \dots \ M]$ and the Gaussian function is $\mathcal{N}(\mathbf{x} | \mu, \sigma) = (1/\sqrt{2\pi}\sigma) \exp(-(x - \mu)^2/2\sigma^2)$. The spectral parameters (f_m, μ_m, σ_m) are the intensity, Doppler shift, and line width at pixel m , respectively. The vector $\mathbf{h}_m^{(a)}$ represents the unscaled Gaussian profile as a column vector shifted by m pixels due to dispersion. This relation can be written in a matrix-vector form as:

$$\mathbf{y}^{(a)} = \begin{bmatrix} | & | & & | \\ \mathbf{h}_1^{(a)} & \mathbf{h}_2^{(a)} & \dots & \mathbf{h}_M^{(a)} \\ | & | & & | \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_M \end{bmatrix} = \mathbf{H}^{(a)}(\boldsymbol{\Theta}) \mathbf{f} \quad (3)$$

where $\boldsymbol{\Theta} = (\mu, \sigma)$. Then, combining measurements from all diffraction orders and also considering the detector noise:

$$\mathbf{y} = \mathbf{H}(\boldsymbol{\Theta}) \mathbf{f} + \mathbf{n} \quad , \quad \boldsymbol{\Theta} = (\mu, \sigma) \quad (4)$$

where $\mathbf{n}$ is the additive white Gaussian noise, the vector $\mathbf{y}$ and the matrix $\mathbf{H}(\boldsymbol{\Theta})$ are formed by vertically stacking $\mathbf{y}^{(a)}$'s and $\mathbf{H}^{(a)}(\mu, \sigma)$'s, respectively. With this parametric formulation, we can regard this inverse problem as a multi-frame semi-blind non-uniform deblurring problem where $\mathbf{f}$ is the sharp image and $\boldsymbol{\Theta}$ are the Gaussian blur parameters at each spatial location. Note that the zeroth diffraction order gives the intensities (i.e. $\mathbf{y}^{(0)} = \mathbf{f} + \mathbf{n}^{(0)}$). Then the goal is to estimate the spatially varying Gaussian blur parameters $\boldsymbol{\Theta}$ and the sharp image $\mathbf{f}$ from the sharp and blurry observations $\mathbf{y}^{(0)}$ and $\mathbf{y}^{(a)}$'s corrupted with noise. Note the nonlinear dependence of measurements on $\boldsymbol{\Theta}$, making this a nonlinear inverse problem.

Combining all the spectral parameters in vector $\mathbf{x}$, where $\mathbf{x} = [\mathbf{f} \ \mu \ \sigma]$ represents the complete set of intensity, Doppler shift, and line width parameters for all spatial locations, the forward model can then be concisely rewritten as $\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$ where $\mathcal{A}(\cdot)$ is the nonlinear forward model.

3. PROPOSED METHOD

3.1. Score-Based Models

Score-based models are generative models that sample from a data distribution $p_{\text{data}}(\mathbf{x})$ using the score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$.

The sampling process starts with Gaussian noise and gradually refines it using the score function to generate samples from the data distribution.

The unified SDE formulation [24] models a forward diffusion process that transforms clean data $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ into Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{max}}^2 \mathbf{I})$:

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt + g(t)d\mathbf{w}_t \quad (5)$$

where $\mathbf{f}(\mathbf{x}_t, t)$ is a drift coefficient, $g(t)$ is the diffusion coefficient, and $\mathbf{w}_t$ is a standard Wiener process. The reverse process samples from the data distribution by running the reverse SDE:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g(t)^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)] dt + g(t)d\bar{\mathbf{w}}_t \quad (6)$$

The score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is approximated by a neural network $\mathbf{s}_\theta(\mathbf{x}_t, t)$ trained using denoising score matching, which leverages the relationship between the score function and the MMSE denoiser via Tweedie's formula [25].

3.2. Diffusion Posterior Sampling (DPS)

For the imaging model $\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$ we need to sample from the posterior $p(\mathbf{x} | \mathbf{y})$. This requires modifying the reverse SDE to use the conditional score $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y})$, which by Bayes rule becomes:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) = \underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)}_{\text{measurement matching term}} \quad (7)$$

The unconditional score comes from the pre-trained model, but the measurement matching term requires approximating the intractable:

$$p_t(\mathbf{y} | \mathbf{x}_t) = \int p(\mathbf{y} | \mathbf{x}_0) p(\mathbf{x}_0 | \mathbf{x}_t) d\mathbf{x}_0 \quad (8)$$

where the likelihood term $p(\mathbf{y} | \mathbf{x}_0)$ is known, but $p(\mathbf{x}_0 | \mathbf{x}_t)$ is unknown.

DPS [22] approximates the problematic $p_t(\mathbf{y} | \mathbf{x}_t)$ term in Eq. (7) with $p(\mathbf{y} | \hat{\mathbf{x}}_0(\mathbf{x}_t))$ which implies that the $p(\mathbf{x}_0 | \mathbf{x}_t)$ term in Eq. (8) is assumed to be $\delta(\mathbf{x}_0 - \hat{\mathbf{x}}_0(\mathbf{x}_t))$ where $\hat{\mathbf{x}}_0(\mathbf{x}_t)$ is the MMSE estimate of $\mathbf{x}_t$ obtained using Tweedie's formula:

$$\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] \approx \hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{\mathbf{x}_t + (1 - \bar{\alpha}_t)\mathbf{s}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \quad (9)$$

With this, the measurement matching term becomes:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \hat{\mathbf{x}}_0(\mathbf{x}_t)) \quad (10)$$

$$\simeq -\frac{1}{\sigma_n^2} \nabla_{\mathbf{x}_t} \|\mathcal{A}(\hat{\mathbf{x}}_0(\mathbf{x}_t)) - \mathbf{y}\|_2^2 \quad (11)$$

which can be plugged back in Eq. (7) to get the conditional score:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) \simeq \mathbf{s}_\theta(\mathbf{x}_t, t) - \frac{1}{\sigma_n^2} \nabla_{\mathbf{x}_t} \|\mathcal{A}(\hat{\mathbf{x}}_0(\mathbf{x}_t)) - \mathbf{y}\|_2^2 \quad (12)$$

Algorithm 1 DPS

Require: N , observation $\mathbf{y}$, step sizes $\{\zeta_i\}_{i=1}^N$, noise levels $\{\tilde{\sigma}_i\}_{i=1}^N$, noise schedule $\{\bar{\alpha}_i, \beta_i\}_{i=1}^N$

- 1: Sample $\mathbf{x}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: **for** $i = N - 1$ to 0 **do**
- 3: $\hat{\mathbf{s}} \leftarrow \mathbf{s}_\theta(\mathbf{x}_i, i)$
- 4: $\hat{\mathbf{x}}_0 \leftarrow \sqrt{\bar{\alpha}_i} \left(\mathbf{x}_i + \frac{1 - \bar{\alpha}_i}{\sqrt{\bar{\alpha}_i}} \hat{\mathbf{s}} \right)$
- 5: Sample $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 6: $\mathbf{x}'_{i-1} \leftarrow \sqrt{\bar{\alpha}_i} \cdot \frac{1 - \bar{\alpha}_{i-1}}{1 - \bar{\alpha}_i} \mathbf{x}_i + \sqrt{\bar{\alpha}_{i-1}} \cdot \frac{\beta_i}{1 - \bar{\alpha}_i} \hat{\mathbf{x}}_0 + \tilde{\sigma}_i \mathbf{z}$
- 7: $\mathbf{x}_{i-1} \leftarrow \mathbf{x}'_{i-1} - \zeta_i \nabla_{\mathbf{x}_i} \|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0)\|_2^2$
- 8: **end for**
- 9: **return** $\hat{\mathbf{x}}_0$

Algorithm 1 shows the DPS sampling procedure where the main difference from unconditional sampling is the gradient descent step in line 7 which drives the samples to match the measurements. We choose the step size $\zeta_i = 1/\|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0(\mathbf{x}_i))\|$.

3.3. DPS for Slitless Spectral Imaging

We adapt the DPS approach for the SSI problem described in Section 2. The score-based model learns to jointly model the intensity, velocity, and line width parameters $\mathbf{x} = [\mathbf{f} \ \boldsymbol{\mu} \ \boldsymbol{\sigma}]^\top \in \mathbb{R}^{3M^2}$ as demonstrated in Figure 2.

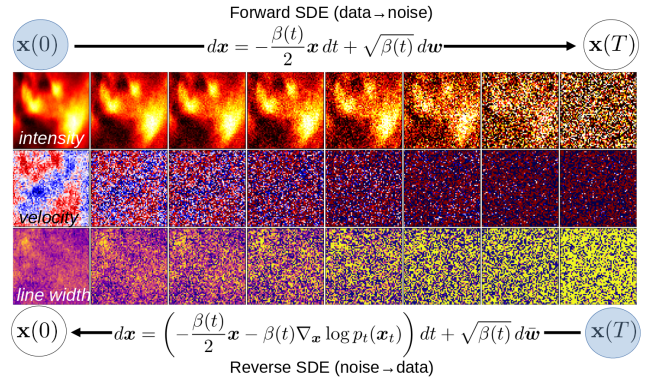


Fig. 2: Illustration of forward and reverse processes for the multi-channel parametrization of the data cube $\mathbf{x} = [\mathbf{f} \ \boldsymbol{\mu} \ \boldsymbol{\sigma}]^\top$

4. NUMERICAL EXPERIMENTS

To evaluate our method, we trained a score-based model using 10,000 samples derived from 1088 scans of Fe XII 195 Å emission observed by the Extreme Ultraviolet Imaging Spectrometer (EIS) aboard the Hinode observatory (see Figure 3). Each sample is a 64×64 three-channel image representing the intensity, Doppler velocity, and spectral line width, obtained by applying Gaussian fits to the calibrated EIS data using the eispac package [26]. Each channel is normalized jointly to the range $[-1, 1]$.

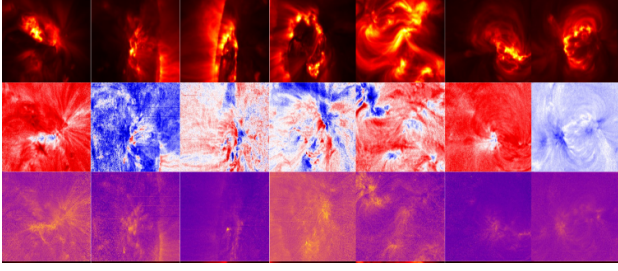


Fig. 3: Samples from the EIS training dataset. Each column shows a sample of x consisting of the intensity, velocity, and line width channels as the three rows, respectively.

We adopt the NCSN++ network architecture in its PyTorch reference implementation, as proposed by Song et al. [24]. We use the cosine noise schedule with default hyperparameters. Training is performed with a batch size of 32 and a learning rate of $5e-6$ for 10,000 epochs, on a single NVIDIA Titan GPU. Total training time is approximately 3 hours.

At test time, we simulate measurements according to $y = \mathcal{A}(x) + n$ using a reserved subset of x samples from the EIS data. Sampling follows the Algorithm 1 procedure with $N = 1000$ total diffusion steps. Reconstruction quality is evaluated using the root-mean-square error (RMSE) between the reconstructed x and the ground truth.

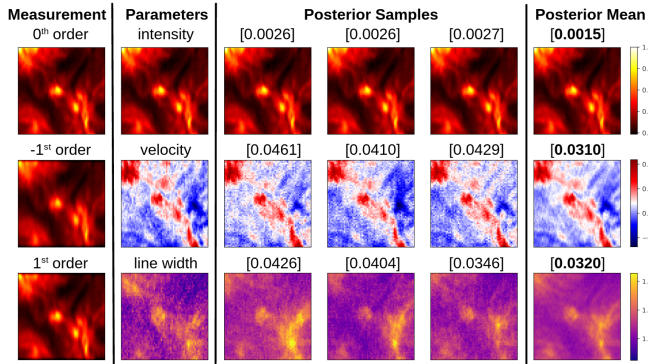


Fig. 4: For a sample measurement, DPS posterior estimates and their mean. The RMSE in pixel units is shown in square brackets for each reconstruction.

Figure 4 shows an example DPS reconstruction of an EIS test measurement with $K = 3$ orders of $\{-1, 0, 1\}$. Three of the 25 posterior samples are shown in the middle columns and the average of them is shown in the last column which approximates the MMSE estimator. There are visible differences between the posterior samples, indicating the usefulness of the posterior sampling approach. Also, as expected, the RMSE of the posterior mean is lower than the individual samples, albeit leading to a blurrier reconstruction.

4.1. Out-of-distribution Experiment

We test the generalization ability of DPS on a synthetically generated spectral cube that looks quite different from EIS

images, and we compare the reconstruction performance with an end-to-end trained U-Net. The methods are compared under low, moderate, and high SNR levels where in all cases we used the U-Net that was trained in the moderate SNR level. The U-Net was trained on the same EIS dataset to perform end-to-end reconstruction from simulated measurements with moderate noise.

Figure 5 shows the reconstructed data cubes by both methods for all SNR levels (decreasing SNR left to right). Although trained on EIS images, DPS is able to reconstruct the parameters well for high and moderate SNR levels, and for the low SNR case, the reconstruction gracefully reverts to a plausible sample drawn from the prior, resembling the natural solar textures it was trained on. The U-net reconstructions have significant errors (middle of I is missing) even in the high SNR case, showing the susceptibility of the end-to-end approaches to distribution shifts.

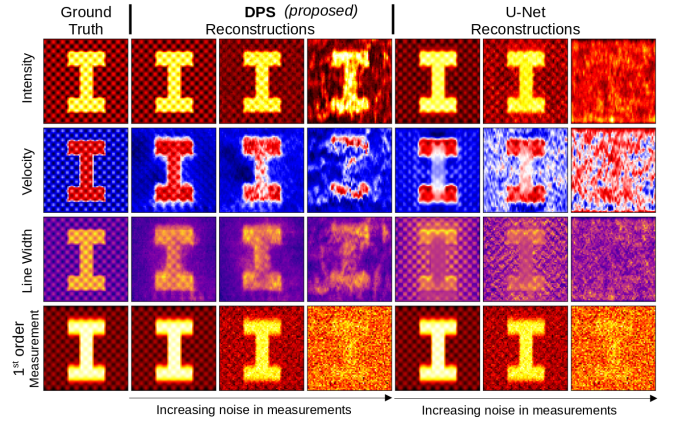


Fig. 5: Reconstruction comparison between U-net and DPS for an out-of-distribution sample under various noise settings.

5. DISCUSSION AND CONCLUSIONS

We adapt a posterior sampling method using score-based models for the slitless spectral image reconstruction problem. The framework generates multiple plausible reconstructions that agree with the measurements, this way it has the advantage of characterizing the ambiguous features in the estimated parameters and quantifying the uncertainties, which are essential considerations for scientific discovery. Another advantage of the method is that it is able to generalize well to out of distribution data where it outperforms the previously developed supervised approach. This is also critical for deep learning based methods in any application of scientific discovery where ground truth data is scarce, and the model needs to generalize to out of training data. Future work could explore scaling this framework to handle multiple spectral lines simultaneously or applying it to other scientific domains facing similar nonlinear inverse problems.

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